

BAIT3153 SOFTWARE PROJECT MANAGEMENT

Radex Company is the largest bus operator in Malaysia operating in Klang Valley areas. Adam, as an operation manager of Radex has been assigned to develop a Cashless Payment System (CPS) for all its stations. The passengers can select the different payment options to pay the fares using credit cards, debit cards methods, touch n go and boost e-wallets. It reduces the queueing time to purchase tickets at the service counter. A budget of RM 150,000 has been approved for this project. The system scope of CPS focuses on the core feature of the graphical user interface design, payment module and integrates with third payment parties. The formation of the project team consists of six staff from the Information Technology department who will be working on this project.

Answer the following questions based on the given scenario.

Question 1

a) Identify and describe **TWO (2)** stakeholders of the CPS project. (6 marks)

1. Passengers (End Users)
 - Passengers are the primary users of the Cashless Payment System. They directly interact with the system to pay bus fares using various methods such as credit/debit cards, Touch 'n Go, and e-wallets. Their needs for convenience, speed, and reliability make them a key stakeholder. The system must ensure usability, security, and efficiency to reduce queue times and improve customer satisfaction.
2. Information Technology Department Staff (Project Team)
 - IT staff are directly responsible for designing, developing, and implementing the CPS. They are crucial stakeholders since they ensure the system meets technical requirements, integrates with third-party payment providers, and remains within the budget. Their expertise directly affects the quality, performance, and success of the project.

- b) Identify and explain **TWO (2)** types of prototyping process models commonly used to address uncertain user requirements and adapt to user changes. (6 marks)

1. Evolutionary Prototyping

- The prototype is built in stages, with each stage adding more features and refinements. It evolves into the final system, making it suitable when requirements are unclear or expected to change, as users can continuously provide feedback.

2. Throw-away Prototyping

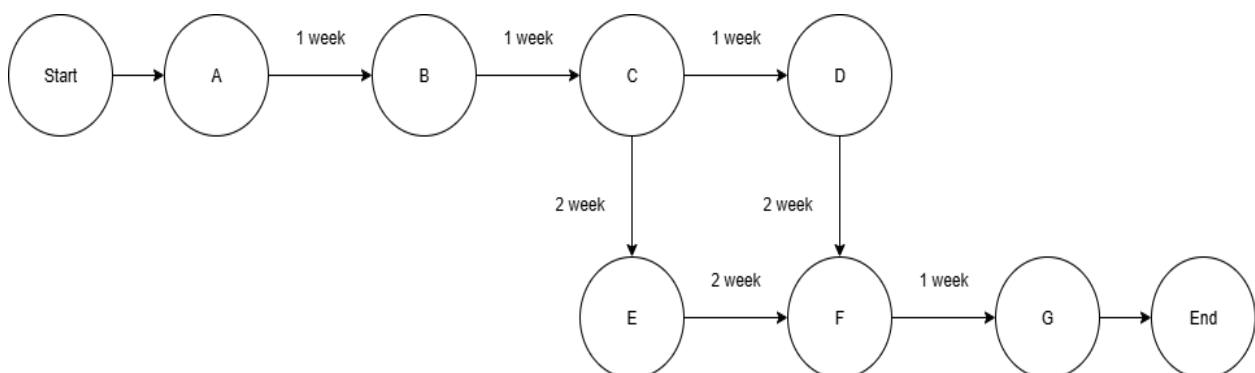
- The prototype is built rapidly to explore ideas and gather user feedback. It is discarded once the necessary information is collected and is not used in the final system, helping to quickly clarify uncertain requirements.

- c) A network diagram can be used to illustrate the interdependencies between tasks in a project. Table 1 shows an example of project tasks, its duration and predecessor.

Table 1: Project tasks, duration and its predecessor

Task ID	Task Name	Duration (in weeks)	Predecessor
A	Project planning	1	None
B	Feasibility study	1	A
C	Refine requirements	1	B
D	Design GUI and database	1	C
E	Design a prototype model	2	C
F	Code for core modules	2	D, E
G	Integration test	1	F

- (i) Create a network diagram based on the task ID, duration and predecessors provided in Table 1. (8 marks)



(ii) Based on the answer given in Question 1 c) (i), identify the critical path based on the task ID and specify the total duration needed to complete the project. (5 marks)

[Total: 25 marks]

The critical path is the one with the longest duration, which is:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow F \rightarrow G$

$= 1 + 1 + 1 + 2 + 1$

$= 6 \text{ weeks}$

Total Duration needed to complete the project: 6 weeks

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Question 2

a) Analyse **TWO (2)** product standards that should be enforced in CPS's project development. Explain your answer. (8 marks)

1. Security Standard (e.g., PCI DSS for Payment Systems)
 - The CPS handles sensitive payment data such as credit cards, debit cards, and e-wallet transactions. Enforcing security standards like PCI DSS (Payment Card Industry Data Security Standard) ensures that payment information is encrypted, stored, and transmitted securely. This reduces risks of fraud, data breaches, and builds passenger trust in using the system.
2. Usability Standard (e.g., ISO 9241 for User Interfaces)
 - Since the CPS will be used daily by passengers at bus stations, the system must be easy to navigate and accessible to all users, including those less familiar with technology. Enforcing usability standards such as ISO 9241 helps ensure that the graphical user interface is intuitive, clear, and efficient, minimizing errors and reducing queuing time. This improves customer satisfaction and adoption of the system.

- b) Given the following information of project team A and B in Table 2.

Table 2: Size-oriented measurement of project team A and team B.

	Non-normalised		Normalised with KLOC	
	Team A	Team B	Team A	Team B
Programming language used	Java	Php	Java	Php
Total Lines of Code (LOC)	206,000	266,000		
Defects found	22	40		
Errors found	25	35		

Note: KLOC is thousand lines of code. Formula used: defects/KLOC and errors/KLOC.

- (i) Calculate the values in Table 2 under normalised with KLOC for both teams. Which project team produces higher quality of work? Explain your answer.
Show your workings in your calculation, round up to five decimal points.

(9 marks)

	Non- normalised		Normalised with KLOC	
	Team A	Team B	Team A	Team B
Programming language used	Java	Php	Java	Php
Total Lines of Code (LOC)	206,000	266,000	$206000/1000$ = 206 KLOC	$266000/1000$ = 266 KLOC
Defects found	22	40	$22/206$ = 0.10680 defects/KLOC	$40/266$ = 0.15038 defects/KLOC
Errors Found	25	35	$25/206$ = 0.12136 errors/KLOC	$35/266$ = 0.13158 errors/KLOC

Team A has lower defects compared to Team B (0.10680 vs 0.15038)

Team A has lower errors compared to Team B (0.12136 vs 0.13158)

Thus, Team A has a higher quality of work than Team B.

- (ii) Evaluate **TWO (2)** drawbacks of size-oriented metric used in CPS's project development.
Explain your answer.

(8 marks)

[Total: 25 marks]

1. Measurement is complex
 - Using size-oriented metrics such as Lines of Code (LOC) oversimplifies software quality and complexity. For the CPS project, this can give misleading results, since a smaller, efficient payment module may appear less productive than a larger, less efficient one.
2. Collecting measures is time-consuming
 - Tracking and maintaining size-based metrics requires additional effort from developers and managers. In CPS development, this diverts valuable time from coding, testing, and improving security, without guaranteeing useful insights.

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Question 3

- a) The risk projection shows the following information:
Adam has identified 50% of the existing software components can be reused and integrated into the current fares payment system. The remaining core modules have to be developed based on new requirements. The project team has to check the existing platform and operating system's version used to integrate with the new modules.
Risk identified is four junior developers have been assigned to develop the 35 new software components with duration of 4 weeks. The average software component is 120 Lines of Code (LOC) and the development cost for each LOC is RM12.00. Risk probability for the identified risk is 50%.
- (i) Calculate the following values using a Halstead's Risk Exposure (RE) metric:
- Cost to develop the 35 software components.
 - Risk exposure.
- Note: show the workings in your calculation. The RE formula is probability X cost.
(3 + 3 marks)

The cost to develop the 35 software components

= No of software components * average LOC per components * cost per LOC

= 35 * 120 * 12

= RM 50,400

Risk exposure, RE

= Risk probability, P * C

= 0.5 * 50,400

= RM 25,200

(ii) Identify **ONE (1)** other potential risk that might occur in the CPS's project. You are required to include the following information:

- Specify the *category* that the risk belongs to.
- Propose a *risk mitigation action* for the identified risk. Explain your answer.

(7 marks)

1. Potential Risk:

- Team members lack experience in critical areas like UI design and cybersecurity

2. Category of Risk:

- Staff size and experience (ST)

3. Risk Mitigation Action:

- Provide targeted training and workshops for the project team, and if necessary, hire or consult with external experts in specialized areas such as UI/UX or security. Since the CPS project involves sensitive payment systems and user interaction design, inexperienced staff may introduce security loopholes or usability issues. By offering training or engaging external expertise, the project team can strengthen its capability, reduce mistakes, and ensure the system is robust and user-friendly.

b) Adam has decided to use the Distributed Version Control (DVC) system for the CPS's project development. Describe **TWO (2)** benefits of DVC system to the project team's collaboration. Provide an explanation in your answers. (6 marks)

1. Backup and Recovery

- Distributed Version Control (DVC) system provides a backup since each developer has a full copy of the repository. If the main repository is corrupted, it can be restored from local copies, ensuring project work continues without major disruption.

2. Offline Development

- Distributed Version Control (DVC) system allows developers to commit changes offline and test the full system locally. This lets them work without a network connection, verify their code before integration, and reduce errors during collaboration.

c) Given below are two software quality attributes:

- Reliability
- Maintainability

In your opinion, which software quality attribute is most important to the CPS's project development? Justify your answer. (6 marks)

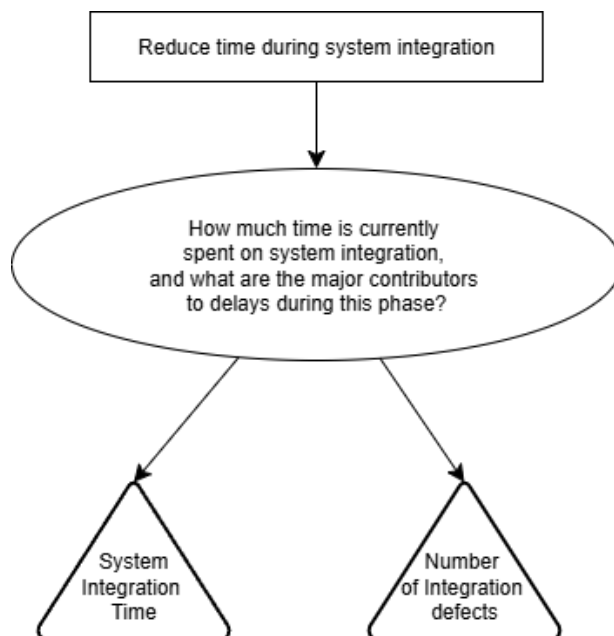
[Total: 25 marks]

For the Cashless Payment System (CPS) project, reliability is the most important software quality attribute. The CPS will be used by thousands of passengers daily for payment. If the system fails, transactions may be lost, payments could be incorrect, directly affecting public trust and business operations. Reliability ensures that the system consistently performs its intended functions without errors or downtime, which is critical for handling real-time financial transactions.

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Question 4

- a) Adam has identified the project goal as *reduce time during system integration* to support the CPS's project process improvement. Create a Goal Question Metric (GQM) proposal for the given goal and include the following information:
- Analyse **ONE (1)** question to assess the given goal.
 - Propose **TWO (2)** relevant software metrics and explain how each of the software metrics can be applied to the suggested question. (9 marks)



1. System Integration Time is the total elapsed time (in hours/days) taken from the beginning of system integration to completion. This metric directly measures the current time spent on integration. By tracking it over multiple projects or iterations, we can observe whether efforts to reduce integration time are effective.
2. Number of Integration Defects is the count of defects identified during the integration phase. High defect counts can significantly delay system integration. Tracking this metric helps identify whether poor quality from earlier phases is increasing integration time. A trend of fewer defects could correlate with reduced integration duration.

b) Given below are two fault management approaches:

- Fault avoidance
- Fault detection

(i) Select a suitable fault management approach that can be applied in the CPS's project development. Explain each of the approaches mentioned above with an example. (9 marks)

Fault Avoidance

- Focuses on preventing faults during development by applying best practices like secure design, code reviews, and formal testing.
- Example: In a cashless payment system, developers can apply encryption standards correctly in the authentication module to prevent future security faults and ensure safe transactions.

Fault Detection

- Identifies faults after they occur using testing, monitoring, or error handling mechanisms.
- Example: In a CPS, real-time monitoring can detect issues like failed API calls or transaction timeouts, enabling prompt alerts and recovery actions..

Fault detection is more suitable for CPS, as it ensures real-time identification and handling of issues like network delays or payment gateway failures, which may not be avoidable during development.

(ii) *Any faults in the delivered software do not cause system errors.* Do you agree with this statement? Provide **TWO (2)** justifications in your answer. (7 marks)

[Total: 25 marks]

Disagree.

1. Faults Can Cause Failures Under Specific Conditions

- A fault in the system may not cause errors during routine usage but can lead to errors when a specific action is taken.
- Example: A small mistake in the code that calculates discounts might work most of the time, but during special promotions, it could give the wrong discount causing either customer loss or business loss.

2. Cashless Systems Handle Sensitive Operations

- Cashless payment systems handle sensitive and real-time financial operations, where even minor software faults can lead to transaction failures, financial loss, or security breaches.
- Example: If there's a fault in the code that checks for duplicate transactions, a user might get charged twice for the same item leading to user complaints and loss of trust.